PROVEIT RESEARCH CONTINUATION

A Cubic-Background Formula for Gamma Normal Forms

Complete mixed polygamma coefficients and Stieltjes–polylogarithm endpoint identities

11 October 2026

Abstract

We determine, at every weight, the complete projection of the inverse-Gamma Bell hierarchy onto functional monomials involving digamma and its first three derivatives. A single entire integral formula gives all coefficients of $u_0^a u_1^k u_2^b u_3^n$ for $n \geq 2$, with explicit parity zeros and a finite two-binomial rational expression. The square case resolves the coupled next-Gamma-sector question in the inspected ProveIt continuation; the cubic and higher powers of u_3 go beyond that question. The proof uses the Euler variational derivative, an injective triangular operator, and four explicit polynomial identities, rather than extrapolation from finite weight calculations. A beta-density formula also replaces the preceding pure-u_2 recurrence by a nonrecursive integral.

The same endpoint-vanishing polynomials generate finite Hurwitz-zeta and Lerch formulas, all-order generalized Stieltjes averages and anchored primitives, and ordinary polylogarithm sums with harmonic endpoint values. Several weighted digamma and first-Stieltjes integrals are evaluated explicitly. Canonical total-derivative primitives are retained when the formal identities are transported to Gamma and harmonic numbers.

Independent exact reductions check 973 mixed coefficients through weight 30; symbolic checks verify the four all-index density identities. Thirty high-precision numerical diagnostics are supplied separately. An arbitrary-derivative-order extension is formulated as an unproved conjecture, with its computational evidence clearly separated. Global priority and exhaustive audit of the incoming archives are not claimed.

Included artifacts. Complete and modular LaTeX sources, compiled PDF, rational verification programs and certificates, numerical diagnostics, source provenance, and repository integration notes.

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1 The research question and the scope of the answer

The inspected ProveIt manuscript develops exact identities among polylogarithms, Gamma derivatives, harmonic tails, and Hurwitz–Stieltjes coordinates [1]. Its differential-polynomial continuation already proves a canonical integration-by-parts normal form and a complete recurrence for the variables u_0, u_1, u_2 [2]. A later continuation treats square sectors on an affine background and explicitly asks for every coefficient

$$[u_0^a u_1^k u_2^b u_3^2] N_{a+2k+3b+8}, \qquad b > 0, \tag{1.1}$$

with parity and the endpoint primitive retained [3, Question 1]. Here u_r represents the rth derivative of digamma, and N_m is the canonical normal representative of $(-1)^m \Gamma(x)(\mathrm{d}/\mathrm{d}x)^m \Gamma(x)^{-1}$.

We answer (1.1) without restricting a, k, b. More strongly, we allow any number $n \geq 2$ of u_3 factors. The answer is a finite rational sum whose summands have four consecutive odd factors in the denominator. Its underlying integral is a beta moment with the endpoint factor $(1-v^2)^3$. This factor is not decorative: its three vanishing endpoint derivatives are exactly what makes the third-order variational transport close.

1.1 A preview in ordinary coefficients

For $m = a + 2k + 3b + 4n$ and $n \geq 2$, the coefficient vanishes when $k + n$ is odd. When $k + n$ is even it is

$$\begin{aligned}&[u_0^a u_1^k u_2^b u_3^n] N_m \\ &\quad = -\frac{m!}{192\, a!\, 2^k k!\, b!\, n!\, 24^b 48^{n-2}} \sum_{j=0}^{b} \sum_{\ell=0}^{n-2} \frac{\binom{b}{j} 3^j \binom{n-2}{\ell}}{\prod_{r=0}^{3} (k+n+2j+2\ell-1+2r)}.\end{aligned} \tag{1.2}$$

For example, the coefficients of $u_2 u_3^2$, $u_2^2 u_3^2$, and $u_1 u_3^3$ are respectively -55, -3640, and $-3185/3$. These are coefficients in the specified rational normal basis, not assertions of arithmetic independence after special-function substitution.

1.2 What is inherited, and what is established here

The normal-form decomposition and the finite-jet Euler principle are inherited. We reproduce their elementary proofs so that the article does not depend on trusting a computation or an inaccessible manuscript. The underlying functional-monomial framework is credited in the source to Guo, Regensburger, and Rosenkranz [4]. The preceding pure-u_2 recurrence is also inherited; its beta-density solution and the complete cubic-background formula are the developments proved below.

The polynomial-weight Hurwitz and Lerch integration-by-parts mechanism uses classical special-function identities [6, 8]. We do not claim that mechanism, the standard Gamma–zeta relations, or elementary partial fractions as new. Their application to the newly determined mixed coefficient kernels gives the exact families and concrete evaluations in Sections 7 and 8. Historical priority for these specializations has not been exhaustively investigated.

The incoming directory was inventoried during source selection, but its binary packages were not exhaustively read or replayed. In particular, the conclusions here are not a claim that every current delivery has been reconciled. The provenance file distinguishes the inspected text from archive names seen in the directory. Since the repository changed during retrieval, we record file-level identifiers where available instead of claiming a single atomic repository snapshot.

1.3 Conventions

All formal polynomial identities are over $\mathbb{Q}$. We write

$$\langle f(v)\rangle = \frac{1}{2}\int_{-1}^{1} f(v)\,\mathrm{d}v, \qquad [t^m]F(t) \text{ for ordinary coefficient extraction.}$$

The exponential generating functions carry their displayed factorials. The notation $\zeta^{(r)}(s,a)$ always means an *order* derivative, $\partial_s^r\zeta(s,a)$; $\psi^{(r)}(a)$ always means an *argument* derivative. Stieltjes constants use the convention

$$\zeta(1+\epsilon,a) = \frac{1}{\epsilon} + \sum_{q\ge 0}\frac{(-1)^q}{q!}\gamma_q(a)\epsilon^q, \qquad \gamma_0(a) = -\psi(a). \tag{1.3}$$

For analytic integral formulas we initially impose $\Re a > 1$, so that $a+v$ stays in the right half-plane. All polylogarithm series are initially used on $|z|<1$, eliminating branch ambiguity. Extensions and removable singularities are stated where used.

2 Canonical normal forms and the Euler principle

Let

$$\mathscr{R} = \mathbb{Q}[u_0,u_1,u_2,\ldots], \qquad \mathsf{D}u_j = u_{j+1}, \qquad \mathrm{wt}(u_j) = j+1.$$

The factor degree of a monomial is its ordinary total degree. A monomial is *functional* if it is a power of u_0, including 1, or if its largest positive derivative index occurs at least twice. Denote the rational span of these monomials by $\mathscr{F}$.

Proposition 2.1 (Canonical decomposition)**.** *Every $P\in\mathscr{R}$ has a unique decomposition*

$$P = \mathsf{N}(P) + \mathsf{D}\mathsf{H}(P), \qquad \mathsf{N}(P)\in\mathscr{F}, \qquad [1]\mathsf{H}(P) = 0. \tag{2.1}$$

Both maps are rational and linear. They preserve factor degree; on a homogeneous input of weight m, $\mathsf{N}(P)$ has weight m and $\mathsf{H}(P)$ has weight $m-1$.

Proof. Suppose that a monomial M has a unique largest derivative u_r, $r\ge 1$. Replace that factor by u_{r-1}, obtaining T, and let c be the multiplicity of u_{r-1} in T. Then

$$M = \frac{1}{c}\mathsf{D}T + \left(M - \frac{1}{c}\mathsf{D}T\right).$$

The remaining monomials have strictly smaller largest derivative index. Repeated application terminates and constructs both maps. The asserted gradings are preserved at every step.

For uniqueness, let a nonconstant H have largest index r. In $\mathsf{D}H$ the term $u_{r+1}\partial_{u_r}H$ is nonzero and consists entirely of monomials with unique largest index $r+1$. It cannot be canceled by the other terms of $\mathsf{D}H$, which have largest index at most r, or by a functional polynomial. Thus $\mathsf{D}H\notin\mathscr{F}$. This proves uniqueness once the constant term of H is fixed. □

The decomposition is a direct sum of vector spaces, not of algebras. In particular, N is not multiplicative: $\mathsf{N}(u_1) = 0$ while $\mathsf{N}(u_1^2) = u_1^2$. The constructive proof gives a complete endpoint primitive, and the supplied rational program implements that proof.

2.1 The inverse-Gamma Bell hierarchy

Define

$$A_0 = 1, \qquad A_{m+1} = u_0 A_m - \mathsf{D}A_m, \tag{2.2}$$

$$\mathcal{A}(t) = \sum_{m\geq 0} A_m \frac{t^m}{m!} = \exp\left(\sum_{r\geq 0} \frac{(-1)^r u_r t^{r+1}}{(r+1)!}\right), \tag{2.3}$$

and put $N_m = \mathsf{N}(A_m)$ and $H_m = \mathsf{H}(A_m)$. Formula (2.3) follows either by differentiating the exponential or by the formal Taylor formula. Under $u_r = \psi^{(r)}(x)$,

$$A_m(x) = (-1)^m \Gamma(x) \frac{\mathrm{d}^m}{\mathrm{d}x^m} \frac{1}{\Gamma(x)}, \qquad \mathcal{A}(t,x) = \frac{\Gamma(x)}{\Gamma(x-t)}. \tag{2.4}$$

These identifications use $\psi = \Gamma'/\Gamma$ and its derivatives [5]. For every fixed m they are ordinary analytic identities away from Gamma poles.

Translation $u_0 \mapsto u_0 + c$ commutes with D and preserves functional monomials. By uniqueness it commutes with N. Consequently $\mathsf{N}\mathcal{A}(t)$ has an Appell factor $e^{u_0 t}$. A second useful identity is

$$\mathsf{N}\mathcal{A}(t) = \mathsf{N}\exp\left(\sum_{j\geq 0} \frac{u_{2j} t^{2j+1}}{2^{2j}(2j+1)!}\right). \tag{2.5}$$

Indeed the exponential before normalization on the right is $e^{t\mathsf{D}/2}\mathcal{A}(t)$. Every coefficient of $(e^{t\mathsf{D}/2}-1)\mathcal{A}(t)$ is a total derivative. The exponent in (2.5) has odd weights only, so every retained monomial has factor degree congruent to its weight modulo two. For $u_0^a u_1^k u_2^b u_3^n$ this says that $k+n$ must be even.

2.2 Euler derivatives and finite jets

Define

$$\mathscr{E}(P) = \sum_{r\geq 0} (-\mathsf{D})^r \partial_{u_r} P.$$

Every sum is finite on a polynomial. The commutator relation $\partial_{u_r}\mathsf{D} = \mathsf{D}\partial_{u_r} + \partial_{u_{r-1}}$, with the last term zero at $r=0$, gives $\mathscr{E}\mathsf{D} = 0$ by cancellation.

Let σ_d set all u_r with $r > d$ to zero and define $\delta_d = \sum_{r=0}^{d-1} u_{r+1}\partial_{u_r}$.

Lemma 2.2 (Finite-jet variational identity)**.** *For every $P \in \mathscr{R}$,*

$$\sigma_d \mathscr{E}(P) = \sum_{r=0}^{d} (-\delta_d)^r \partial_{u_r}(\sigma_d \mathsf{N}(P)). \tag{2.6}$$

Furthermore,

$$\mathscr{E}\mathcal{A}(t) = \int_0^t e^{s\mathsf{D}}\mathcal{A}(t)\,\mathrm{d}s. \tag{2.7}$$

The integral in (2.7) is coefficientwise formal.

Proof. Since $\mathscr{E}\mathsf{D} = 0$, replace P by $\mathsf{N}(P)$ on the left of (2.6). A normal monomial removed by σ_d has at least two factors of its highest index, which exceeds d. One partial derivative removes at most one such factor; subsequent applications of D preserve a factor of index greater than d. Hence its

specialization vanishes. For retained monomials, the ideal generated by $u_{d+1}, u_{d+2}, \dots$ is D-stable, and specialization changes D to δ_d. This proves (2.6).

By (2.3), $\partial_{u_r}\mathcal{A}(t) = (-1)^r t^{r+1}\mathcal{A}(t)/(r+1)!$. Therefore

$$\mathscr{E}\mathcal{A}(t) = \sum_{r\ge 0} \frac{t^{r+1}}{(r+1)!}\mathsf{D}^r\mathcal{A}(t),$$

which is the formal integral in (2.7). □

Normal reduction must precede truncation. For example, $\sigma_1(u_0u_2) = 0$, whereas $\sigma_1\mathsf{N}(u_0u_2) = -u_1^2$. The finite-jet lemma, not commutation of these operations, is the reason one can obtain a closed low-derivative equation.

3 A beta-density solution of the quadratic background

We use the dimensionless variables and polynomials

$$y = \frac{u_1t^2}{2}, \quad z = u_2t^3, \quad w = u_3t^4, \qquad h(v) = \frac{1+3v^2}{24}, \quad k(v) = \frac{v+v^3}{48}, \tag{3.1}$$

with $P_r(v) = (1-v^2)^r$. In particular $h' = v/4$ and $k' = h/2$. Write

$$\sum_{m\ge 0}\sigma_2 N_m\frac{t^m}{m!} = e^{u_0t}F_0(y,z), \qquad F_0(y,z) = \sum_{b\ge 0} z^b f_b(y).$$

The previous continuation gives a differential equation and rational recurrence for these functions [2]. We solve that recurrence by a single nonrecursive beta density.

Theorem 3.1 (Pure-u_2 density)**.** *Let*

$$f_{\mathrm{aff}}(y) = \frac{1}{2}\left(\cosh y + \frac{\sinh y}{y} - y\,\mathrm{Shi}(y)\right), \qquad \mathrm{Shi}(y) = \int_0^y \frac{\sinh v}{v}\,\mathrm{d}v. \tag{3.2}$$

All singularities displayed at zero are removable. Then $f_0 = f_{\mathrm{aff}}$, $f_1 = 0$, and, for every $b \ge 2$,

$$\boxed{f_b(y) = \frac{1}{192\,b!}\left\langle P_2(v)h(v)^{b-2}e^{yv}\right\rangle.} \tag{3.3}$$

Equivalently, with $\mathcal{E}_2(x) = (e^x - 1 - x)/x^2$ and $\mathcal{E}_2(0) = 1/2$,

$$F_0(y,z) = f_{\mathrm{aff}}(y) + \frac{z^2}{192}\left\langle P_2(v)e^{yv}\mathcal{E}_2(zh(v))\right\rangle. \tag{3.4}$$

This is entire jointly in y, z, and

$$\partial_z^2 F_0(y,z) = \frac{1}{192}\left\langle P_2(v)e^{yv+zh(v)}\right\rangle. \tag{3.5}$$

Proof. Let $A = 2y + (z/2)\partial_y$. Lemma 2.2, specialized to the quadratic background, gives

$$F_0 - \frac{1}{2}A\partial_y F_0 + A^2\partial_z F_0 = \left\langle e^{yv+zh(v)}\right\rangle. \tag{3.6}$$

To see the right-hand side, set $s = t(1+v)/2$ in (2.7); the exponent after specialization is $u_0 t + yv + zh(v)$. The two sides of (3.6) are the Euler identity divided by $te^{u_0 t}$.

Since a single u_2 is not functional, $f_1 = 0$. Coefficient z^0 in (3.6) yields $f_0 - yf_0' = \sinh y/y$. The unique solution with $f_0(0) = 1$ and no linear term is (3.2). In particular,

$$f''_{\mathrm{aff}}(y) = -\frac{1}{2}\left\langle (1-v^2)e^{yv} \right\rangle. \tag{3.7}$$

For $n \geq 1$, the remaining coefficients in the PDE give

$$4(n+1)y^2 f_{n+1} = \frac{1}{n!}\left\langle h^n e^{yv} \right\rangle - (n+1)f_n - (2n-1)yf_n' - \frac{n-2}{4}f''_{n-1}. \tag{3.8}$$

The case $n = 1$, using (3.7), gives $f_2 = \langle P_2 e^{yv} \rangle /384$. For instance, this follows immediately from $P_2'' = 4(3v^2 - 1)$ by integrating twice by parts. The endpoint values of P_2 and P_2' are zero.

For $n \geq 2$ the elementary polynomial identity

$$\begin{aligned} 4(P_2 h^{n-1})'' = {} & 192h^n - (n+1)P_2 h^{n-2} + (2n-1)(vP_2 h^{n-2})' \\ & - \frac{n(n-2)}{4}v^2 P_2 h^{n-3} \end{aligned} \tag{3.9}$$

proves that (3.3) satisfies (3.8). At $n = 2$ the last term is understood as zero. For all $n \geq 3$, divide by h^{n-3}, differentiate with $h' = v/4$, and substitute $24h = 1 + 3v^2$; both sides become the same polynomial in n and v. This is a direct all-index algebraic identity, not an inference from checking finitely many n.

To pass from (3.9) to (3.8), integrate against e^{yv} and transfer one or two v derivatives. Endpoint terms vanish because of P_2. The terms involving v and v^2 give f_n' and f''_{n-1}. After multiplication by $192n!$, the result is exactly the recurrence. Multiplication by y^2 is injective on formal power series, so the initial values determine the sequence uniquely. This proves the density formula. Summing z^b gives (3.4), and differentiating twice gives (3.5). Compact integration of entire functions proves joint entire convergence. □

For even k, extracting the coefficient of y^k gives a particularly simple rational replacement for the source recurrence:

$$k![y^k]f_b(y) = \frac{1}{24^{b-1}b!}\sum_{j=0}^{b-2}\frac{\binom{b-2}{j}3^j}{(k+2j+1)(k+2j+3)(k+2j+5)}. \tag{3.10}$$

For odd k it is zero. Formula (3.10) follows by expanding $(1+3v^2)^{b-2}$ and using $\int_0^1 v^K(1-v^2)^2\,\mathrm{d}v = 8/((K+1)(K+3)(K+5))$.

4 The complete cubic-background formula

We now determine the entire projection with highest retained derivative index three. The words "cubic background" refer to the fact that a formal shift of u_0 is a cubic polynomial when $u_r = 0$ for $r > 3$. They do not restrict the power of u_3 in the answer.

Theorem 4.1 (Complete cubic normal form)**.** *With the notation of* (3.1)*,*

$$\sum_{m\ge 0} \sigma_3 N_m \frac{t^m}{m!} = e^{u_0 t} F(y, z, w), \tag{4.1}$$

where

$$\boxed{\begin{aligned} F(y,z,w) = f_{\mathrm{aff}}(y) + \frac{z^2}{192}\,\langle P_2 e^{yv}\mathcal{E}_2(zh)\rangle \\ - \frac{w^2}{9216}\left\langle P_3 e^{yv+zh}\mathcal{E}_2(wk)\right\rangle. \end{aligned}} \tag{4.2}$$

This function is entire jointly in y, z, w. Writing $F = \sum_{n\ge 0} w^n F_n(y, z)$, its coefficients are F_0 from Theorem 3.1, $F_1 = 0$, and

$$F_n(y,z) = -\frac{1}{9216\,n!}\left\langle P_3(v)k(v)^{n-2}e^{yv+zh(v)}\right\rangle, \qquad n \ge 2. \tag{4.3}$$

4.1 The closed Euler equation

Set

$$A = 2y + \frac{z}{2}\partial_y, \qquad B = \partial_z, \qquad C = \frac{1}{2}\partial_y, \qquad L = A + wB.$$

On a function $e^{u_0 t}G(y, z, w)$ the derivation δ_3 acts as $e^{u_0 t}t^{-1}LG$. Lemma 2.2 therefore gives

$$F - \frac{1}{2}LF_y + L^2F_z - L^3F_w = \left\langle e^{yv+zh(v)+wk(v)}\right\rangle. \tag{4.4}$$

In detail, after $s = t(1 + v)/2$ in (2.7), the exponent of the shifted Bell exponential is

$$u_0 t + \frac{u_1 t^2 v}{2} + \frac{u_2 t^3(1 + 3v^2)}{24} + \frac{u_3 t^4(v + v^3)}{48}.$$

This establishes the right-hand side of (4.4) with all normalizations fixed.

The commutators $[B, A] = C$ and $[C, A] = 1$ give

$$\begin{aligned} L^2 &= A^2 + w(2AB + C) + w^2B^2, \\ L^3 &= A^3 + w(3A^2B + 3AC + 1) + w^2(3AB^2 + 3CB) + w^3B^3. \end{aligned}$$

Put

$$R_n(y,z) = \frac{1}{n!}\left\langle k(v)^n e^{yv+zh(v)}\right\rangle.$$

Comparison of w^n in (4.4) gives, for $n \ge 1$,

$$\begin{aligned} (n+1)A^3F_{n+1} = (1-n)F_n - \frac{1+3n}{2}A\partial_y F_n + (1-3n)A^2\partial_z F_n \\ + (5-3n)A\partial_z^2 F_{n-1} + \frac{3(1-n)}{2}\partial_y\partial_z F_{n-1} \\ + (3-n)\partial_z^3 F_{n-2} - R_n. \end{aligned} \tag{4.5}$$

Negative subscripts mean zero. The w^0 equation is exactly the quadratic equation because the normal basis excludes a single u_3.

Lemma 4.2 (Uniqueness of the cubic hierarchy)**.** *The operator A^3 is injective on $\mathbb{C}[[y,z]]$. Consequently (4.5), together with the specified F_0 and $F_1 = 0$, determines every F_n uniquely.*

Proof. If G is nonzero, let j be its least nonzero z-degree and write that coefficient as $g(y)$. The coefficient of z^j in A^3G is $8y^3g(y)$, which is nonzero. Indeed every use of $(z/2)\partial_y$ instead of $2y$ raises z-degree. This proves injectivity and then the successive uniqueness assertion. □

4.2 Why the beta density solves every recurrence

Let $E(v) = e^{yv+zh(v)}$. The key transport relation is

$$AE(v) = 2\partial_v E(v), \qquad A^j \langle Q(v)E(v)\rangle = (-2)^j \left\langle Q^{(j)}(v)E(v)\right\rangle. \tag{4.6}$$

The second equality holds when $Q, Q', \dots, Q^{(j-1)}$ vanish at both endpoints. For Q divisible by P_3, it applies for $j \le 3$. The coefficients of A are independent of v, so iteration of the first equality and ordinary integration by parts prove the second.

Proof of Theorem 4.1. We verify that (4.3) satisfies (4.5). Uniqueness then identifies it with the actual normal projection, so no unproved assumption about a formal solution is needed.

For $n = 1$, the recurrence reduces to

$$2A^3F_2 = 2A\partial_z^2 F_0 - R_1.$$

Using (3.5) and (4.6), this is the polynomial identity

$$\frac{P_3'''}{1152} = -\frac{P_2'}{48} - k. \tag{4.7}$$

For $n = 2$, the required identity is

$$-8(P_3k)''' = -P_3 + 7(vP_3)' - 20(hP_3)'' - 96P_2h + 9216k^2. \tag{4.8}$$

Both follow by differentiating the displayed polynomials.

For every $n \ge 3$, substituting (4.3) in the recurrence, multiplying by $-9216n!$, and using (4.6) reduces the assertion to

$$\begin{aligned} -8(P_3k^{n-1})''' = {} & (1-n)P_3k^{n-2} + (1+3n)(vP_3k^{n-2})' \\ & + 4(1-3n)(hP_3k^{n-2})'' - 2n(5-3n)(h^2P_3k^{n-3})' \\ & + \frac{3}{2}n(1-n)vhP_3k^{n-3} + n(n-1)(3-n)h^3P_3k^{n-4} \\ & + 9216k^n. \end{aligned} \tag{4.9}$$

At $n = 3$ the term with factor $3 - n$ is zero; no negative power is part of the asserted identity. For $n \ge 4$ all terms are polynomials.

Here is a finite algebraic way to verify (4.9) for an indeterminate n, making the all-index step explicit. Define

$$\mathcal{G}_{a,q}(U) = U' + a\frac{q'}{q}U.$$

On the open set where $k \neq 0$, divide (4.9) by k^{n-4} and replace its left side by $-8\mathcal{G}^3_{n-4,k}(P_3k^3)$. The terms on its right become, in order,

$$\begin{aligned}
&(1-n)P_3k^2, && (1+3n)\mathcal{G}_{n-4,k}(vP_3k^2),\\
&4(1-3n)\mathcal{G}^2_{n-4,k}(hP_3k^2), && -2n(5-3n)\mathcal{G}_{n-4,k}(h^2P_3k),\\
&\tfrac{3}{2}n(1-n)vhP_3k, && n(n-1)(3-n)h^3P_3, \qquad 9216k^4.
\end{aligned}$$

Substitute $P_3 = (1-v^2)^3$, $24h = 1+3v^2$, and $48k = v+v^3$. Expanding the derivatives and collecting powers of n makes the difference identically zero in $\mathbb{Q}(v)[n]$. Only three derivatives are needed, independently of n. Clearing denominators proves the identity on the entire polynomial domain. The same calculation at $n = 3$, or its polynomial continuation, proves that case with the stated zero-term convention. The supplied script records these exact symbolic operations.

All integrations by parts in this verification have zero endpoint terms: the worst derivative transfer is of order three, and its density has the factor P_3. The exceptional terms involving F_0 occur only at $n = 1, 2$ and were handled separately. Thus the recurrence is verified at every index. Lemma 4.2 proves (4.3). Finally, $\sum_{n\geq 2} w^n k^{n-2}/n! = w^2\mathcal{E}_2(wk)$ gives (4.2). The integrands are entire and uniformly bounded on compact parameter sets and the compact interval, proving joint entirety. □

Remark 4.3 (Proof status of the algebraic step). The four identities (3.9), (4.7), (4.8), and (4.9) are explicit finite polynomial or rational-function identities. Their symbolic verification treats n as an indeterminate. The separate checks at weights at most 30 are useful regression tests but are not the reason that the theorem holds for every weight.

5 Finite coefficient formulas and explicit mixed families

Corollary 5.1 (All mixed u_3 coefficients)**.** *Let $a, k, b \geq 0$, $n \geq 2$, and $m = a+2k+3b+4n$. If $k+n$ is odd, then $[u_0^a u_1^k u_2^b u_3^n]N_m = 0$. If $k+n$ is even, then*

$$\begin{aligned}
&[u_0^a u_1^k u_2^b u_3^n]N_m\\
&= -\frac{m!}{9216\,a!2^k k!b!n!24^b 48^{n-2}} \int_0^1 v^{k+n-2}(1-v^2)^3(1+3v^2)^b(1+v^2)^{n-2}\,\mathrm{d}v.
\end{aligned} \tag{5.1}$$

Equivalently, the finite rational formula (1.2) holds. Every coefficient not forced to vanish by parity is strictly negative.

Proof. Expand $e^{u_0 t}$, e^{yv}, $e^{zh(v)}$, and $\mathcal{E}_2(wk(v))$ in (4.2). The conversion from y^k to u_1^k contributes 2^{-k}, and conversion to N_m contributes $m!$. The remaining integrand is odd precisely when $k+n$ is odd; otherwise its average on $[-1,1]$ is its integral on $[0,1]$. This proves (5.1). Expand the two binomials and use

$$\int_0^1 v^K(1-v^2)^3\,\mathrm{d}v = \frac{48}{(K+1)(K+3)(K+5)(K+7)}, \qquad K > -1. \tag{5.2}$$

Formula (5.2) follows by $x = v^2$ and three elementary integrations of the beta integral, or by expanding $(1-v^2)^3$. Here $K = k+n-2+2j+2\ell \geq 0$, so every denominator is positive. This proves the finite formula and the sign assertion. □

5.1 The source square-sector question

For $n = 2$ and even k, (1.2) becomes

$$\boxed{\begin{aligned}&[u_0^a u_1^k u_2^b u_3^2] N_{a+2k+3b+8} \\ &= -\frac{(a+2k+3b+8)!}{384\, a!2^k k! b! 24^b} \sum_{j=0}^{b} \frac{\binom{b}{j} 3^j}{(k+2j+1)(k+2j+3)(k+2j+5)(k+2j+7)}.\end{aligned}} \tag{5.3}$$

It is zero for odd k. Thus every $b > 0$ in the specific question (1.1) is covered; no weight-dependent linear system remains to be solved.

Define the normalized square coefficient for even k by

$$g_b(k) = k![y^k z^b w^2] F(y, z, w).$$

Then

$$g_b(k) = -\frac{1}{18432 b! 24^b} \int_0^1 v^k (1-v^2)^3 (1+3v^2)^b \, \mathrm{d}v. \tag{5.4}$$

For example,

$$g_0(k) = -\frac{1}{384(k+1)(k+3)(k+5)(k+7)}, \tag{5.5}$$

$$g_1(k) = -\frac{1}{2304(k+1)(k+5)(k+7)(k+9)}, \tag{5.6}$$

$$g_2(k) = -\frac{(k+2)(k+6)}{27648 \prod_{j=0}^{5}(k+2j+1)}, \tag{5.7}$$

$$g_3(k) = -\frac{k^3+15k^2+62k+63}{497664 \prod_{j=0}^{6}(k+2j+1)}. \tag{5.8}$$

These rational functions are asserted as coefficient formulas at even nonnegative integers k. Their analytic continuation in k is also provided by the beta integral for $\Re k > -1$, but the coefficient at an odd power of y remains zero.

5.2 Nonlinear sectors beyond the square

The complete formula also gives genuinely coupled cubic and quartic sectors. A small sample, with no u_0 factor, is

Weight	Functional monomial	Coefficient in N_m
8	u_3^2	-1
11	$u_2 u_3^2$	-55
12	$u_1^2 u_3^2$	-165
14	$u_2^2 u_3^2$	-3640
14	$u_1 u_3^3$	$-3185/3$
16	u_3^4	$-10325/3$
17	$u_1 u_2 u_3^3$	$-1023400/3$
19	$u_2 u_3^4$	-1639225

All coefficients with an additional power u_0^a follow immediately from the Appell factorial ratio. The rational denominators in this basis need not be integral, even though the Bell polynomial before normal reduction has integral coefficients.

The formulas also reveal a useful structural distinction. The pure highest-index-two sector has positive beta density, whereas the entire highest-index-three sector has negative beta density, subject to parity. This observation motivates the all-order conjecture in Section 9.3; it does not by itself establish that conjecture.

6 Gamma identities, harmonic values, and retained primitives

The normal-form theorem is an identity in a free differential algebra. Its analytic meaning must keep both the primitive in (2.1) and the distinction between a projection and the full normal polynomial.

Theorem 6.1 (Transport with the exact endpoint)**.** *Let $0 < A < B$, $m \geq 0$, and write $U(x) = (\psi(x), \psi'(x), \ldots)$. Then*

$$\begin{aligned}\int_A^B N_m(U(x))\,\mathrm{d}x = (-1)^m \int_A^B \Gamma(x)\frac{\mathrm{d}^m}{\mathrm{d}x^m}\frac{1}{\Gamma(x)}\,\mathrm{d}x \\ - H_m(U(B)) + H_m(U(A)).\end{aligned} \tag{6.1}$$

The polynomial H_m is computed by the terminating rational construction in Proposition 2.1. In particular, the formula gives an anchored primitive for every normal combination N_m without suppressing an integration-by-parts boundary constant.

Proof. Substitute $u_r = \psi^{(r)}(x)$ into $A_m = N_m + \mathsf{D}H_m$. Under this substitution D is ordinary argument differentiation. Use (2.4) and integrate over the compact interval. All functions are analytic there. □

For use with the compact cubic formula, let $R_m = N_m - \sigma_3 N_m$. The precise projected identity is therefore

$$\begin{aligned}\int_A^B (\sigma_3 N_m)(U(x))\,\mathrm{d}x = (-1)^m \int_A^B \Gamma(x)(\Gamma^{-1})^{(m)}(x)\,\mathrm{d}x \\ - [H_m(U(x))]_A^B - \int_A^B R_m(U(x))\,\mathrm{d}x.\end{aligned} \tag{6.2}$$

One must not erase the last term when the higher-derivative sector is nonzero. For $m \leq 9$ it vanishes by weight, but it need not vanish from weight ten onward.

6.1 Two explicit normal polynomials

The weight-eight identity is

$$N_8 = u_0^8 - 70u_0^4u_1^2 + 28u_0^2u_2^2 - 7u_1^4 - u_3^2. \tag{6.3}$$

For this case the complete endpoint polynomial is

$$\begin{aligned}H_8 = &-4u_0^7 + 56u_0^5u_1 - 70u_0^4u_2 - 140u_0^3u_1^2 + 56u_0^3u_3 \\ &+ 252u_0^2u_1u_2 - 28u_0^2u_4 + 112u_0u_1^3 - 112u_0u_1u_3 \\ &- 84u_0u_2^2 + 8u_0u_5 - 98u_1^2u_2 + 20u_1u_4 + 36u_2u_3 - u_6.\end{aligned} \tag{6.4}$$

Direct differentiation verifies $A_8 = N_8 + \mathsf{D}H_8$. Equations (6.1), (6.3), and (6.4) are an ordinary finite-interval integral identity involving digamma, trigamma, and higher polygammas, with an entirely explicit endpoint.

At weight eleven,

$$\begin{aligned} N_{11} = u_0^{11} - 330u_0^7u_1^2 + 462u_0^5u_2^2 - 1155u_0^3u_1^4 - 165u_0^3u_3^2 \\ + 550u_0^2u_2^3 + 990u_0u_1^2u_2^2 + 11u_0u_4^2 - 55u_2u_3^2. \end{aligned} \tag{6.5}$$

Thus $R_{11} = 11u_0u_4^2$. This gives a concrete test of why the projection alone cannot be substituted for the full N_{11} in an unqualified Gamma integral. The coefficient -55 is the first coupled case of (5.3). The full H_{11} and other exact primitives are supplied as rational JSON certificates.

6.2 Generalized harmonic numbers

Let $H_N^{(r)} = \sum_{j=1}^N j^{-r}$, $H_N = H_N^{(1)}$, and let γ denote Euler's constant. The standard shift formulas give

$$u_0 = H_N - \gamma, \qquad u_r = (-1)^{r+1}r!(\zeta(r+1) - H_N^{(r+1)}), \quad r \geq 1, \tag{6.6}$$

at $x = N + 1$ [5, 6]. Let U_N denote this tuple. Then

$$N_m(U_N) + (\mathsf{D}H_m)(U_N) = m![t^m]\frac{\Gamma(N+1)}{\Gamma(N+1-t)}. \tag{6.7}$$

The right side also equals

$$m![t^m]\exp\left((H_N - \gamma)t - \sum_{r\geq 2}\frac{\zeta(r) - H_N^{(r)}}{r}\,t^r\right). \tag{6.8}$$

This follows by substituting (6.6) into (2.3). It is initially a local analytic series in t and an identity of every finite coefficient. The derivative $\mathsf{D}H_m$ is formed in the free differential algebra *before* substitution; it is not a finite difference in N.

Consequently (5.3) and the nonlinear coefficients specify exact mixed harmonic-tail contributions to every identity (6.7). This is a systematic family of identities, not a claim that the individual harmonic products have been shown arithmetically independent or that every integral of one isolated polygamma monomial has been separately evaluated.

7 Hurwitz endpoints and all Stieltjes orders

The new coefficient densities naturally supply the polynomial family

$$W_{b,n}(v) = (1-v^2)^3(1+3v^2)^b(v+v^3)^{n-2}, \qquad b \geq 0,\ n \geq 2. \tag{7.1}$$

Multiplication by v^k gives all coefficient moments in (5.1). Every such polynomial and its first two derivatives vanish at $v = \pm 1$. This creates a finite endpoint calculus starting with the third derivative, uniformly in b, n, k.

7.1 The polynomial endpoint identity

Let P be any polynomial of degree d and define

$$M_P(s;a) = \frac{1}{2}\int_{-1}^{1} P(v)\zeta(s,a+v)\,\mathrm{d}v, \qquad \mu_P = \langle P(v)\rangle\,.$$

Use $s-1^{\underline{j+1}} = (s-1)(s-2)\cdots(s-j-1)$.

Proposition 7.1 (Finite Hurwitz endpoint formula)**.** *For $\Re a > 1$, as an identity of meromorphic functions of s,*

$$M_P(s;a) = -\frac{1}{2}\sum_{j=0}^{d} \frac{\left[P^{(j)}(v)\zeta(s-j-1,a+v)\right]_{-1}^{1}}{s-1^{\underline{j+1}}}. \tag{7.2}$$

Its only possible pole is at $s=1$, with residue μ_P. All other apparent singularities of the right side are removable after summation. If P, P', P'' vanish at both endpoints, the sum starts at $j=3$.

Proof. The classical argument derivative $\partial_a\zeta(s-1,a) = -(s-1)\zeta(s,a)$ [6] gives the first integration by parts. Repeat until $P^{(d+1)} = 0$. Each endpoint term has the minus sign shown in (7.2). This proves the formula away from the exceptional integers. On the compact interval and locally uniformly for $\Re a > 1$, the integrand is analytic in s except for its simple pole at one. Its residue integrates to μ_P. Meromorphic continuation therefore proves the endpoint identity and all stated cancellations. □

This proposition is the standard polynomial integration-by-parts mechanism. The specialization to $W_{b,n}$ is what connects the newly computed Gamma normal coefficients to spectral special functions. No endpoint regularization in the variable v is needed: the integrals are ordinary compact-interval integrals. Laurent coefficients in s are a different operation.

7.2 Finite harmonic coefficients for every Stieltjes order

For $j \geq 0$ define $\eta_{j,\ell}$ by

$$\prod_{r=1}^{j}(1-\epsilon/r)^{-1} = \sum_{\ell\geq 0}\eta_{j,\ell}\epsilon^{\ell} = \exp\left(\sum_{r\geq 1}\frac{H_j^{(r)}}{r}\epsilon^r\right). \tag{7.3}$$

The empty product at $j=0$ is one. In particular,

$$\eta_{j,0} = 1, \qquad \eta_{j,1} = H_j, \qquad \eta_{j,2} = \frac{H_j^2 + H_j^{(2)}}{2}.$$

These are finite polynomials in generalized harmonic numbers at each fixed ℓ.

Theorem 7.2 (All-order Stieltjes averages)**.** *For $q \geq 0$ and $\Re a > 1$,*

$$\begin{aligned}\langle P(v)\gamma_q(a+v)\rangle = \frac{(-1)^{q+1}q!}{2}\sum_{j=0}^{d}\frac{(-1)^j}{j!} \\ \times\left[P^{(j)}(v)\sum_{r=0}^{q+1}\eta_{j,q+1-r}\frac{\zeta^{(r)}(-j,a+v)}{r!}\right]_{-1}^{1}.\end{aligned} \tag{7.4}$$

The formula is finite at every q and applies in particular to every polynomial $v^k W_{b,n}$.

Proof. At $s = 1 + \epsilon$,

$$\frac{1}{\epsilon^{\underline{j+1}}} = \frac{(-1)^j}{j!\epsilon} \sum_{\ell \geq 0} \eta_{j,\ell} \epsilon^\ell. \tag{7.5}$$

Expand $\zeta(-j + \epsilon, a + v)$ in its order variable and insert (7.5) into (7.2). Coefficient ϵ^q is the right side of (7.4) multiplied by $(-1)^q/q!$. On the integral side the same coefficient is $(-1)^q \langle P\gamma_q \rangle / q!$ by (1.3). Uniformity on compact parameter sets justifies coefficient extraction under the integral. The coefficient of ϵ^{-1} is the separately retained residue μ_P; it must not be discarded before multiplying the expansions. □

Corollary 7.3 (Anchored primitives and argument derivatives)**.** *For a_0, a in the half-plane $\Re a > 1$, with the integral taken along any path in that half-plane,*

$$\int_{a_0}^{a} \langle P(v)\gamma_q(c + v)\rangle \, \mathrm{d}c = \frac{(-1)^{q+1}}{q+1} \partial_s^{q+1} \big(M_P(s; a) - M_P(s; a_0)\big)\Big|_{s=0}. \tag{7.6}$$

For $r \geq 1$, let $e_j^{[r]}$ be the elementary symmetric polynomial of degree j in $1, 1/2, \ldots, 1/r$. Then

$$\partial_a^r \langle P(v)\gamma_q(a + v)\rangle = (-1)^{q+r} q! r! \sum_{j=0}^{\min(q,r)} e_j^{[r]} \frac{\partial_s^{q-j} M_P(s; a)\big|_{s=r+1}}{(q-j)!}. \tag{7.7}$$

Here the derivative ∂_s on $M_P(r + 1; a)$ is evaluated in its first argument. Both formulas reduce to finite endpoint expressions by Proposition 7.1 and its removable limits.

Proof. From $\partial_a \zeta(s, a) = -s\zeta(s + 1, a)$ and the Laurent expansion,

$$\partial_a \zeta^{(q+1)}(0, a) = (-1)^{q+1}(q + 1)\gamma_q(a).$$

Integrate in a and average against P to obtain (7.6); the displayed subtraction fixes the anchor exactly. For the derivative formula, use

$$\partial_a^r \zeta(s, a) = (-1)^r (s)_r \zeta(s + r, a), \qquad (1 + \epsilon)_r = r! \sum_{j=0}^{r} e_j^{[r]} \epsilon^j,$$

and compare coefficients at $s = 1 + \epsilon$. □

For $q = 0$, the primitive is the weighted log-Gamma difference because $\zeta'(0, a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi)$ [6]. Formula (7.7) at $q = 0$ recovers the weighted polygamma integrals, using $\psi^{(r)}(a) = (-1)^{r+1} r! \zeta(r + 1, a)$.

7.3 Explicit digamma and Stieltjes evaluations

Corollary 7.4 (Four concrete evaluations)**.** *All integrals below are ordinary integrals over* $[-1,1]$*:*

$$\int_{-1}^{1} P_3(v)\psi(2+v)\,\mathrm{d}v = 16\zeta'(-3) + 12\zeta'(-5) + \frac{2}{7}, \tag{7.8}$$

$$\int_{-1}^{1} P_3(v)(1+3v^2)\psi(2+v)\,\mathrm{d}v = 64\zeta'(-3) + 240\zeta'(-5) + 48\zeta'(-7) + \frac{547}{1800}, \tag{7.9}$$

$$\int_{-1}^{1} P_3(v)(v+v^3)\psi(2+v)\,\mathrm{d}v = \frac{32}{105} - \frac{84\zeta(5)}{\pi^4} + \frac{1575\zeta(7)}{2\pi^6} - \frac{2835\zeta(9)}{2\pi^8}, \tag{7.10}$$

$$\int_{-1}^{1} P_3(v)\gamma_1(2+v)\,\mathrm{d}v = 8\zeta''(-3) + 6\zeta''(-5) + \frac{88}{3}\zeta'(-3) + \frac{137}{5}\zeta'(-5) - \frac{16}{15}\log 2 + \frac{521}{900}. \tag{7.11}$$

Proof. At $a = 2$ the endpoint arguments are 1 and 3. Use the exact shift

$$\zeta(s,3) = \zeta(s) - 1 - 2^{-s}. \tag{7.12}$$

For $q = 0$, Theorem 7.2 and $\gamma_0 = -\psi$ give

$$\int_{-1}^{1} P(v)\psi(2+v)\,\mathrm{d}v = \sum_{j=0}^{d} \frac{(-1)^j}{j!}\left[P^{(j)}(v)\big(\zeta'(-j,2+v) + H_j\zeta(-j,2+v)\big)\right]_{-1}^{1}. \tag{7.13}$$

The nonzero derivatives at $v = 1$ needed for the three polynomials are listed below; their derivatives at -1 follow from parity.

j	$P_3^{(j)}(1)$	$(P_3(1+3v^2))^{(j)}(1)$	$(P_3(v+v^3))^{(j)}(1)$
3	-48	-192	-96
4	-288	-2304	-1344
5	-720	-14400	-10080
6	-720	-54720	-50400
7	0	-120960	-171360
8	0	-120960	-362880
9	0	0	-362880

Substituting these finite lists into (7.13) and using (7.12) proves (7.8) and (7.9). The third polynomial first gives

$$-112\zeta'(-4) - 140\zeta'(-6) - 18\zeta'(-8) + \frac{32}{105}.$$

The zeta functional equation implies $\zeta'(-2r) = (-1)^r(2r)!\zeta(2r+1)/(2(2\pi)^{2r})$; substitution gives (7.10).

Finally use $q = 1$ in (7.4). Its inner sum is

$$\tfrac{1}{2}\zeta''(-j,2+v) + H_j\zeta'(-j,2+v) + \tfrac{1}{2}(H_j^2 + H_j^{(2)})\zeta(-j,2+v).$$

With the first derivative column and (7.12), collecting terms yields (7.11). These calculations require no numerical recognition of constants. □

The second and third integrals use precisely the first mixed square and nonlinear cubic coefficient kernels, apart from their fixed scaling constants. The theorem thus links the algebraic coefficients to explicit zeta and Stieltjes periods. It does not assert that these particular low-degree evaluations have no earlier appearance in the literature.

8 Polylogarithm sums and Lerch order derivatives

The beta moments in the mixed coefficient formula can be summed directly. This gives ordinary polylogarithms, not only order derivatives or formal spectral symbols. We then give a second family involving Lerch and polylogarithm order derivatives.

8.1 An exact polylogarithm sum for every square kernel

Write

$$P_b(v) = (1-v^2)^3(1+3v^2)^b = \sum_{j=0}^{b+3} d_{b,j} v^{2j}, \qquad c_b = \frac{1}{18432\, b!\, 24^b}. \tag{8.1}$$

Then (5.4) gives

$$g_b(2\ell) = -c_b \sum_{j=0}^{b+3} \frac{d_{b,j}}{2\ell + 2j + 1}. \tag{8.2}$$

The elementary odd-part polylogarithm is

$$\chi_r(z) = \frac{\mathrm{Li}_r(z) - \mathrm{Li}_r(-z)}{2} = \sum_{\ell \geq 0} \frac{z^{2\ell+1}}{(2\ell+1)^r}, \qquad |z| < 1.$$

The defining polylogarithm series and its analytic conventions are standard [7].

Theorem 8.1 (All coefficient-weighted odd polylogarithm sums)**.** *Let $b \geq 0$, $m \geq 1$, and $|z| < 1$. Then*

$$\sum_{\ell \geq 0} \frac{g_b(2\ell) z^{2\ell+1}}{(2\ell+1)^m} = -c_b \sum_{j=0}^{b+3} d_{b,j} T_{m,j}(z), \tag{8.3}$$

where $T_{m,0}(z) = \chi_{m+1}(z)$ and, for $j \geq 1$,

$$\begin{aligned} T_{m,j}(z) = {} & \sum_{r=1}^{m} \frac{(-1)^{m-r}}{(2j)^{m-r+1}} \chi_r(z) \\ & + \frac{(-1)^m z^{-2j}}{(2j)^m} \left(\chi_1(z) - \sum_{r=0}^{j-1} \frac{z^{2r+1}}{2r+1} \right). \end{aligned} \tag{8.4}$$

The apparent negative powers at $z = 0$ are removable.

Proof. Substitution of (8.2) reduces the assertion to evaluating

$$T_{m,j}(z) = \sum_{\ell \geq 0} \frac{z^{2\ell+1}}{(2\ell+1)^m (2\ell + 2j + 1)}.$$

For $j = 0$ this is χ_{m+1}. For $j \geq 1$, use the finite partial fraction identity, with $x = 2\ell + 1$ and $c = 2j$,

$$\frac{1}{x^m (x+c)} = \sum_{r=1}^{m} \frac{(-1)^{m-r}}{c^{m-r+1} x^r} + \frac{(-1)^m}{c^m (x+c)}. \tag{8.5}$$

Sum the terms using absolute convergence. In the last term, shift the odd index by $2j$, obtaining (8.4). The bracket starts with z^{2j+1}, proving removability at zero. □

Corollary 8.2 (Zeta and harmonic endpoint values)**.** *The limit at $z = 1$ is given by the same finite sum (8.3), with*

$$T_{m,0}(1) = (1 - 2^{-m-1})\zeta(m+1), \tag{8.6}$$

$$T_{m,j}(1) = \sum_{r=2}^{m} \frac{(-1)^{m-r}(1-2^{-r})\zeta(r)}{(2j)^{m-r+1}} + \frac{(-1)^{m+1}}{(2j)^m}\left(H_{2j} - \frac{1}{2}H_j\right), \qquad j \geq 1. \tag{8.7}$$

An empty sum in (8.7) is zero.

Proof. Combine the two $\chi_1(z)$ terms in (8.4) before letting $z \to 1$. Their coefficient is a constant multiple of $1 - z^{-2j}$; its product with the logarithmic singularity tends to zero. For $r \geq 2$, $\chi_r(1) = (1-2^{-r})\zeta(r)$, and the finite odd harmonic sum is $H_{2j} - H_j/2$. The defining series for $T_{m,j}$ converges absolutely at one. So does the coefficient-weighted series; indeed the beta formula gives $g_b(2\ell) = O(\ell^{-4})$ for fixed b. These observations justify passage to the endpoint. □

For example, a base normalization consequence is

$$\sum_{\ell \geq 0} \frac{1}{(2\ell+1)^2(2\ell+3)(2\ell+5)(2\ell+7)} = \frac{\pi^2}{384} - \frac{17}{1080}. \tag{8.8}$$

This example is an elementary partial-fraction evaluation. The uniform formula (8.3) is useful because the coefficients g_b have already been identified with every mixed square sector, including the previously unanswered $b > 0$ family.

Termwise application of $z\partial_z$ lowers m, and integration against $\mathrm{d}z/z$ from zero raises m. Absolute convergence on compact subdisks justifies both operations. Thus (8.3) also provides anchored antiderivatives and derivative identities for the same coefficient families, without introducing new undetermined constants.

8.2 Lerch transforms and polylogarithm order derivatives

For $|z| < 1$ and $\Re a > 1$, set

$$L_P(z, s; a) = \frac{1}{2}\int_{-1}^{1} P(v)\Phi(z, s, a+v)\,\mathrm{d}v, \qquad \Phi(z, s, a) = \sum_{n \geq 0} \frac{z^n}{(n+a)^s}.$$

The Lerch series is entire in s on this disk, and $\partial_a \Phi(z, s-1, a) = -(s-1)\Phi(z, s, a)$ [8]. Hence the same proof as Proposition 7.1 gives

$$L_P(z, s; a) = -\frac{1}{2}\sum_{j=0}^{d} \frac{\left[P^{(j)}(v)\Phi(z, s-j-1, a+v)\right]_{-1}^{1}}{s - 1^{\underline{j+1}}}. \tag{8.9}$$

Unlike the Hurwitz case, *all* the apparent poles of this expression, including $s = 1$, are removable after the finite sum is combined.

Proposition 8.3 (Order-one Lerch average)**.** *For $|z| < 1$ and $\Re a > 1$,*

$$L_P(z, 1; a) = -\frac{1}{2}\sum_{j=0}^{d} \frac{(-1)^j}{j!}\left[P^{(j)}(v)\big(\partial_s \Phi(z, -j, a+v) + H_j \Phi(z, -j, a+v)\big)\right]_{-1}^{1}. \tag{8.10}$$

At $a = 2$ this is a finite expression in $\mathrm{Li}_{-j}(z)$, $\partial_s \mathrm{Li}_s(z)|_{s=-j}$, and elementary terms.

Proof. Expand (8.9) at $s = 1 + \epsilon$ using (7.5). Entirety in s forces the coefficient of ϵ^{-1} to cancel. The constant coefficient is exactly (8.10). At the integer endpoints,

$$\Phi(z, s, 1) = z^{-1} \operatorname{Li}_s(z), \qquad \Phi(z, s, 3) = z^{-3}\big(\operatorname{Li}_s(z) - z - z^2 2^{-s}\big). \tag{8.11}$$

These follow by shifting the defining series and prove the final claim. □

For an entirely explicit implementation, put $A_j(z) = P^{(j)}(1)z^{-3} - P^{(j)}(-1)z^{-1}$ and write $\dot{\operatorname{Li}}_{-j}(z) = \partial_s \operatorname{Li}_s(z)|_{s=-j}$. The endpoint expression inside the summation in (8.10), at $a = 2$, is

$$\begin{aligned} &A_j(z)\big(\dot{\operatorname{Li}}_{-j}(z) + H_j \operatorname{Li}_{-j}(z)\big) \\ &\quad + P^{(j)}(1)\left(2^j z^{-1} \log 2 - H_j(z^{-2} + 2^j z^{-1})\right). \end{aligned} \tag{8.12}$$

The apparent singularities at $z = 0$ cancel by (8.11). Formula (8.10) therefore connects every polynomial $v^k W_{b,n}$ to a finite family of polylogarithm order derivatives with no infinite reduction step.

The limits $z \to 1$ and $s \to 1$ should not be interchanged without analysis: the Lerch transform is entire in s for $|z| < 1$, whereas its Hurwitz counterpart can have the genuine residue μ_P at $s = 1$. The direct Hurwitz formula in Theorem 7.2 keeps that residue and is the appropriate normalization for Stieltjes constants.

9 Verification, source audit, and further research

9.1 What has actually been verified

The exact verification program constructs the Bell polynomials by (2.2) and reduces them by the independent terminating algorithm of Proposition 2.1. It then compares the resulting rational coefficients with the beta formulas. This route does not use the cubic variational recurrence to compute the normal coefficients.

The supplied recorded run has the following scope:

Exact check	Count
Full identities $A_m = N_m + \mathsf{D}H_m$, weights 0–30	31
Functional support assertions in those normal forms	2974
Cubic-sector coefficients, including prescribed parity zeros	973
Quadratic-sector coefficients	575
Affine-sector coefficients	256
All-index symbolic density identities	4
Independent beta versus double-sum comparisons	216
Endpoint-vanishing assertions for the mixed kernels	288

The four symbolic identities involve an indeterminate n where appropriate; the other counts are finite regression checks. A second exact script checks all four displayed spectral integrals, the four low-b rational coefficient families, twelve partial-fraction instances, and the printed H_8 and N_{11} against the certificates. The exact run used Python 3.13.5 and SymPy 1.14.0. Rational normal-form coefficients use Python's exact `Fraction` arithmetic. JSON certificates include both N_m and H_m at weights 8, 11, 14, 20, 24, and 30, and a separate file lists nonzero mixed examples.

The numerical program performs 30 independent diagnostics: eight small-parameter comparisons of the compact integral with independently reduced normal coefficients, three weighted digamma

evaluations, nine Hurwitz endpoint identities, nine polylogarithm sums, and one first-Stieltjes integral. Most use 50 decimal digits; the Stieltjes integral uses 35. The configured tolerances are 10^{-38} and 10^{-28} respectively, and all tests pass. These are arbitrary-precision floating-point diagnostics, not certified error intervals. A truncated normal-series comparison is likewise a diagnostic, not a bound for its uncomputed remainder.

No proof assistant was run. The article supplies analytic and algebraic proofs, the exact script checks the finite rational identities, and the numerical script tests the special-function normalizations. These are three different forms of evidence and are not conflated.

9.2 Corrections and safeguards from the source review

Within the narrowly inspected mathematical core, no substantive error is asserted. The normal decomposition, the finite-jet lemma, and the prior quadratic recurrence are retained with attribution. The new density solution is a continuation, not a purported correction to those results.

There is an editorial duplication in the inspected manuscript README: the paragraph beginning with the complete 1,032-input first-letter X family, its rank 1,008, and kernel dimension 24 appears twice consecutively [1]. The proposed edit is simply to remove the second occurrence, after checking that the current version still has it. No mathematical assertion in that paragraph is changed.

The following normalization safeguards are important for integration. They are not claims that the inspected source made each error. First, $\sigma_d \mathsf{N} \neq \mathsf{N}\sigma_d$ in general; the example after Lemma 2.2 supplies an exact counterexample. Second, $\sigma_3 N_m$ is not the full normal polynomial once higher derivative sectors occur, as (6.5) shows. Third, the polynomial primitive H_m is essential in ordinary definite integrals. Fourth, the Laurent residue at $s = 1$ must be retained when multiplying the endpoint expansion; otherwise the generalized harmonic corrections in (7.4) can be lost. Finally, the finite rational coefficient rank of a declared basis says nothing by itself about arithmetic independence of the corresponding zeta periods.

The incoming archive names were visible, including packages concerning Gamma normalizations, colored moments, and Stieltjes resonances. Their contents were not exhaustively reconciled with this article. The integration note therefore requests a focused overlap check rather than claiming that this is the globally newest result in every one of those packages. The repository remains unchanged by this delivery.

9.3 An all-derivative-order conjecture

The density formula has a striking extension which is supported by exact checks but is not proved here. For $j \geq 0$, put

$$q_j(v) = \frac{(1+v)^{j+1} - (v-1)^{j+1}}{2^{j+1}(j+1)!}. \tag{9.1}$$

Then $q_0 = 1$, $q_1 = v/2$, $q_2 = h$, $q_3 = k$, and $q_j' = q_{j-1}/2$ for $j \geq 1$.

Conjecture 9.1 (Universal highest-derivative density)**.** *Let $r \geq 1$ and let $M = u_0^a \prod_{j=1}^r u_j^{m_j}$ be functional with $m_r \geq 2$ and weight $W = a + \sum_{j=1}^r (j+1)m_j$. Then*

$$\begin{aligned}[M]N_W = {} & \frac{(-1)^r W!}{4^r r!(r+1)!\, a! \prod_{j=1}^r m_j!} \\ & \times \left\langle (1-v^2)^r q_r(v)^{m_r-2} \prod_{j=1}^{r-1} q_j(v)^{m_j} \right\rangle.\end{aligned} \tag{9.2}$$

The cases $r = 1, 2, 3$ follow from the affine formula, Theorem 3.1, and Theorem 4.1. The exact exploratory program compares 438 nonzero coefficients with highest index at least four through weight 24, and every one agrees. That finite comparison does *not* prove the conjecture, nor does it check all absent monomials beyond the proved sectors.

In generating-function form the conjectured contribution of the largest index r is

$$\frac{(-1)^r x_r^2}{4^r r!(r+1)!} \left\langle (1-v^2)^r \exp\left(\sum_{j=1}^{r-1} x_j q_j(v)\right) \mathcal{E}_2(x_r q_r(v)) \right\rangle, \qquad x_j = u_j t^{j+1}, \tag{9.3}$$

with the overall Appell factor $e^{u_0 t}$. For $r = 1$ this expression is the affine correction $f_{\mathrm{aff}}(y) - 1$ with $x_1 = 2y$. The factor $(1-v^2)^r$, the derivative ladder for q_j, and the normal-basis condition all point toward a general variational transport proof. Such a proof is not supplied by the cubic calculation alone.

9.4 Further research questions

1. Prove or refute the universal density. Establish (9.2) at arbitrary r, or exhibit a first counterexample. A desirable proof would replace the order-three polynomial identity by a uniform identity using $q_j' = q_{j-1}/2$ and the r endpoint zeros of $(1-v^2)^r$. The conjecture is specific enough for independent symbolic or proof-assistant attack.

2. Find a closed endpoint formula, not only a terminating algorithm. The canonical H_m is completely specified and verifiable at every weight, but a compact generating representation analogous to (4.2) would make analytic integrations more transparent. It must respect the constant normalization and distinguish a primitive of the full normal form from one of a truncated projection.

3. Determine the arithmetic image of the new mixed kernels. For rational $a > 1$, classify the finite Hurwitz-zeta derivative spaces actually occupied by $\left\langle v^k W_{b,n}(v)\gamma_q(a+v) \right\rangle$. The endpoint formula gives an explicit spanning family, but does not prove minimality or numerical period independence. Distribution identities and parity may yield substantially smaller exact families.

4. Reconcile Lerch confluence with the Hurwitz residue. Develop a uniform two-parameter formula near $(z, s) = (1, 1)$ that separates the emerging Hurwitz pole from the analytic Lerch remainder. This would turn the present warning about order of limits into an exact confluence theorem for every mixed kernel and its order derivatives.

5. Evaluate additional nonlinear polygamma combinations. The formulas here supply exact coefficient identities and retained Gamma endpoints. A further task is to evaluate selected individual integrals, for example those involving $\psi''(\psi''')^2$, in explicitly declared convergent zeta-derivative or multiple-polylogarithm coordinates. The high-derivative remainder in (6.2) must remain visible until it is independently reduced.

6. Study convergence beyond a fixed projection. Every fixed cubic projection is entire in its finite set of variables. This does not establish convergence of a sum over all derivative indices after substituting the infinite jet of digamma. A rigorous study should separate formal coefficient identities from analytic convergence of the full normal-form generating series.

7. Formalize the proof-carrying coefficient compiler. Formalize the terminating monomial reduction and the identity $A_m = N_m + \mathsf{D}H_m$, then verify the rational certificate files. The all-index density identities are a compact subsequent target because they reduce to polynomial arithmetic with an indeterminate index.

8. Preserve the separate period problems. The present work does not resolve the general unequal-even-tail optimal prefix problem, the Gaussian S_6 identity, or the revised S_8 candidate in the inspected manuscript. Their exact branch conventions and lower weight constants remain separate research tasks. No conclusion about those problems follows from the coefficient formula proved here.

Conclusion

A previously coupled infinite coefficient problem collapses to one entire beta-density formula. The endpoint factor that proves the formula also organizes exact Hurwitz, Stieltjes, Lerch, and ordinary polylogarithm identities. The result is an all-weight theorem in a specified normal basis, with retained primitives and reproducible checks, rather than a finite table promoted to a conjectural theorem. The universal extension has been deliberately left at its actual status: a precise and strongly tested further question.

References

[1] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*, manuscript README, VladimirReshetnikov/ProveIt, inspected 11 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/Polylogarithms/docs/manuscript. File-level provenance is recorded in the accompanying manifest.

[2] ProveIt research continuation, *A variational equation for all Gamma coefficients of derivative order at most two*, section 04 of *Harmonic Tail Gamma*, `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/harmonic-tail-gamma/sections/04_gamma_sectors.tex`, VladimirReshetnikov/ProveIt, inspected 11 October 2026. Contains the normal-form construction, finite-jet Euler identity, and the pure-u_2 recurrence used here.

[3] ProveIt research continuation, *Further research questions and concrete next targets*, section 11 of *Affine Gamma Polarization*, `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/affine-gamma-polarization/sections/11_questions.tex`, VladimirReshetnikov/ProveIt, inspected 11 October 2026. Question 1 is the coupled square-sector problem answered in (5.3).

[4] L. Guo, G. Regensburger, and M. Rosenkranz, *On integro-differential algebras*, Journal of Pure and Applied Algebra **218** (2014), 456–473. https://arxiv.org/abs/1212.0266. DOI: 10.1016/j.jpaa.2013.06.015. The source continuation attributes the functional-monomial framework to Proposition 4.10; the elementary decomposition needed here is reproved in Proposition 2.1.

[5] NIST Digital Library of Mathematical Functions, Chapter 5, especially §5.15, *Polygamma Functions*. https://dlmf.nist.gov/5.15. Consulted for standard Gamma and polygamma conventions.

[6] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*. https://dlmf.nist.gov/25.11. In particular, the shift, argument-derivative, special-value, and Gamma–zeta identities.

[7] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*. https://dlmf.nist.gov/25.12.

[8] NIST Digital Library of Mathematical Functions, §25.14, *Lerch's Transcendent*. https://dlmf.nist.gov/25.14.